

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 32)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

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0 1 . 2

Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 1 . 3

In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 1 . 4

Explain how each of the following reduces NO_x emissions.

[3 marks]

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Fluidised bed combustion

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Catalytic converters

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Urea sprays

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10

0 2 Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 2 . **1** Define the term **environmental half-life**.

[1 mark]

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0 2 . **2** Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

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0 2 . 3

Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

0 2 . 4

Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

0 2 . 5

Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

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2

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0 3

Table 1 shows some techniques used to locate mineral deposits.

0 3 . 1

Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

03 . 2

Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

03 . 3

Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

03 . 4

Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

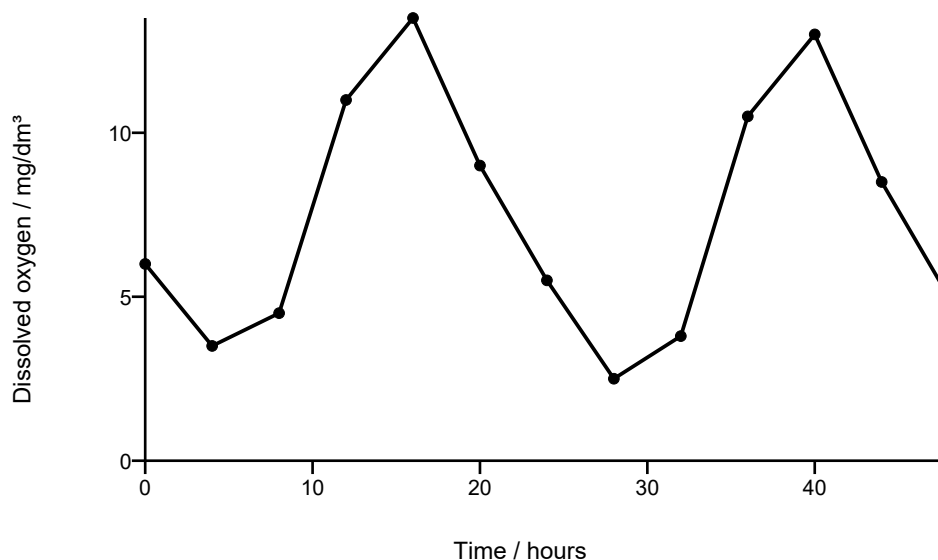
[1 mark]

10

0 4

Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 4

. 1

Describe the pattern shown in Figure 1.

[2 marks]

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0 4

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Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 4 . 3

Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

0 4 . 4

Give **two** other harmful effects of algal blooms.

[2 marks]

1

2

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0 5

A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 5 . 1

Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

05 . 2

Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using $D = N(N-1) / \sum n(n-1)$. Show your working.

[3 marks]**Answer**

05 . 3

The ungrazed meadow had $D = 1.8$ with the same number of species. Explain what this shows.

[2 marks]

05 . 4

Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

05 . 5

Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

10

0 6

Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 6**1**

Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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0 6**2**

Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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0 6**3**

Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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06 . 4

Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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07 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

- A Ozone is destroyed only by CFCs ☐
- B Ozone absorbs mainly infrared radiation ☐
- C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐
- D Ozone concentrations are highest at ground level ☐

07 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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07 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0	8
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The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s ⁻¹	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0	8
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1

Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer

times

0	8
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2

Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

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08 . 3

Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

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08 . 4

Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

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10

09 . 1

Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

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09 . 2

Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

09 . 3

Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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09 . 4

Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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10 . 1

Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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Extra space

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10 . **2** State what is meant by the **residence time** of a substance in a reservoir. **[1 mark]**

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

OR

1 1 . 2

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

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END OF QUESTIONS